

(A01) ERNE Fittings GmbH Hauptstraße 48 6824 Schlins Austria ernefittings the safe connection	(A02) INSPECTION CERTIFICATE 3.1 EN10204:2004	(A03) No.: 2024-000607-SO Rev.: 0 Date: 2024-01-15 Page: 1 / 4
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(A06)

Customer Erne Fittings USA, Inc. 520 Post Oak Boulevard Suite 710 TX 77027 Houston United States	Consignee Maverick International-South Warehouse 16095 Gallier Road 77622 Hamshire, Texas United States
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Order

(A07, A08)

Erne order no.	001018TJ
Erne order pos.	0021
Customer order no.	6363
Order date	2023-03-20

Requirements of the order

(B01, B02, B03, B04)

tee MSS SP-75-19 + CSA Z245.11-17 material . . . : WPHY 60/WPHY 65/Gr. 448 Cat-II TR-fitting . . . : MSS SP-75-2019 . . . : CSA Z245.11-17 TR-addit. fitting : NACE MR0175-2021/ISO15156:2020 . . . : NACE MR0103/ISO17945:2015 . . . : 13.1 & 15 . . . : For WPHY65: ASTM A860-22 certificate . . . : EN 10204:2004/3.1 rust protection. : Millstandard execution. . . : Seamless

Item specification

(B01)

Manufacturer item no.	144877
Product form	tees
Article designation	R-S-WPHY60/65/448CatII-16"/8"-XS/XS
Steel grade	WPHY_60/WPHY_65/Gr_448_Cat-II

Product dimensions (nominal)

(B09, B10, B11)

Diameter	406.40 mm
Diameter 2	219.07 mm
Wall thickness	12.70 mm
Wall thickness 2	12.70 mm

Product

(B04, B07, B08)

Code No.	5FMRT
Quantity	6 pcs
Manufacturing process	Hot formed

Marking, Identification

© AUSTRIA Z245.11-22 448 M46C SS A860 WPHY60/WPHY65 QT 5FMRT 16"/8" XS/XS MT ©

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Pre material (C70)

Heat no.	Starting material
894866	Seamless tubes

Heat Analysis (C71)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
894866	0.140 %	0.21 %	1.26 %	0.013 %	0.004 %	0.14 %	0.05 %	0.09 %	0.24 %	0.06 %

Heat no.	Nb	Al	Ti	N	CE _{IIW}
894866	0.00 %	0.039 %	0.00 %	0.01 %	0.42

Product Analysis (C81)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
894866	0.170 %	0.22 %	1.28 %	0.010 %	0.003 %	0.12 %	0.05 %	0.08 %	0.21 %	0.05 %

Heat no.	Nb	Al	B	Ti	N	CE _{IIW}	CE _N
894866	0.00 %	0.035 %	0.0002 %	0.00 %	0.00 %	0.45	0.44

Formula legend (C71)

Code	Name	Formula
CE _{IIW}	Carbon Equivalent (IIW)	$C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$
CE _N	Carbon Equivalent (N)	$C+(0.75+0.25*\tanh(20*(C-0.12)))*(Si/24 + Mn/6 + Cu/15 + Ni/20 + (Cr+Mo+Nb+V)/5 + 5B)$

Heat treatment (B04)

Process	Temperature	Cooling
Quenched and tempered		
	920 °C	Cooling in water
	640 °C	Cooling in still air

Hardness test (C30, C31)

Test report no.	Method	Min	Max
19028	HB	-	190

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Tensile test

(C01, C02, C03, C10, C11, C12, C13)

Sample no.	Sample orientation	Sample size	Temperature	Yield Strength	Tensile strength	Elongation
				R _{p0.2}		
84X_028	Transversal	Round specimen Ø 8.88 mm L: 35.60 mm 4D	+23 °C	73000 PSI	88500 PSI	26 %

Notched bar impact test

(C40, C41, C42, C43, C44)

Sample no.	Sample orientation	Temperature	Dimension of test sample			Work	Lat. width	Shear fracture
			bo	So	Notch shape			
84X_028	Longitudinal	-46 °C	10.0 mm	80.0 mm ²	V-Notch	274.0 J 281.0 J 278.0 J Ø 277.7 J	2.56 mm 2.57 mm 2.55 mm	- - -
84X_034	Transversal	-46 °C	7.5 mm	60.0 mm ²	V-Notch	102.0 J 111.0 J 109.0 J Ø 107.3 J	- - -	60.0 % 60.0 % 60.0 %

Additional tests

(D02, D51)

Test name	Test report no.	Test standard	Conditions	Rate	Result
Visual inspection and dimensional check					Without objection

Statement of Compliance

(Z01)

The products were manufactured, sampled, tested and inspected in accordance with the specifications and were found to meet the requirements.

Inspection representative

(A05, Z02, Z03)

Date	2024-01-15
Validated by	Inspection Representative
Name / Signature / Stamp	Sommer Stefanie
Telephone	+43 5524 501 328
Mail	stefanie.sommer@ernefittings.com

The certificate is issued by the computerized system and is valid without signature.

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Indication in parentheses correspond to attributes according to EN 10168

Official approvals

(Z05)

Approved by	Name	Certificate No.
TÜV SÜD Industrie Service GmbH (Notified Body 0036)	AD 2000 Merkblatt W0; Pressure Equipment Directive (PED) 2014/68/EU Annex I, Paragraph 4.3 ; EN 764-5, Para. 4.2.	DGR-0036-QS-W 129/2002/MUC-01
TÜV SÜD Management Service GmbH	EN ISO 9001:2015	12 100 3054 TMS

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In order to compare this document with the original document you have to scan the QR code. If you have doubts about the authenticity do not hesitate to contact us by email: check.certificates@ernefittings.com

